
Final Project Report: Flashlight Power Converter Design

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1 Introduction and Project Motivation

For the final project, a discrete, low-voltage boost LED driver is designed and built for a Cree XHP50.3 6 V LED (XHP50D-00-0000-0D0UH240G). (*Xlamp Xhp50.3 Leds* 2025) The objective is to realize a compact, thermally-managed flashlight that is bright enough to illuminate a workspace or driveway while remaining efficient and comfortable to hold. Although high-power LED flashlights are widely available, most commercial designs conceal the internal power-converter behavior and provide little visibility into regulation, transient response, or thermal performance. This work instead emphasizes a transparent, measurement-friendly architecture in which the power electronics are explicitly designed, modeled, and probed, rather than relying on off-the-shelf drivers.

The project builds on prior hardware developed over the summer, where a custom STM32-based board was designed to manage a Li-ion cell, deliver up to 3 A on a power-path rail using the BQ2125, and interface with an onboard accelerometer to implement a “shake-to-wake” LED control gesture. During that effort, several commercial boost LED drivers, including the TPS61169 and TPS61500, were evaluated and found unable to deliver sufficient current to fully drive the Cree XHP50.3. These limitations motivated the development of a higher-power, constant-current architecture in which a boost converter measures LED current via a shunt resistor and regulates it directly to achieve stable brightness and predictable thermal behavior. In the present work, the prototype operates from a single 3 V rail representing a fresh pair of alkaline cells, with no upstream regulation. The converter must boost from 3 V to approximately 6–6.5 V to drive the Cree XHP50.3 in its 6 V configuration (typical forward voltage 5.6–6.2 V at 1.4 A, 3 A continuous rating), while maintaining acceptable component stress and temperature margins. (*Xlamp Xhp50.3 Leds* 2025)

For the 6.2222 graduate extension, the flashlight driver is used as a laboratory platform for converter modeling, control design, and energy-flow analysis. The implementation uses discrete analog building blocks (LM358 op-amps, comparator, 22 μ H inductor, current shunt), alongside small-signal modeling and MATLAB state-space simulation to study loop stability and transient response. Time-domain and frequency-domain measurements (inductor current, switch node voltage, LED current, control signals) will be used to validate the models. In addition, silicon MOSFET and GaN FET devices will be compared in the same boost topology to quantify differences in switching loss and thermal performance. Tellegen’s theorem will be applied to track energy flow through the converter and verify that measured power losses are consistent with theoretical predictions. Finally, a small switched-capacitor voltage doubler will

be constructed to generate short, higher-voltage flashes on the LED, extending Lab 3 concepts to examine how brief high-voltage bursts affect brightness and heating.

2 Independent Inquiry Background and Control-Theoretic Context

The independent inquiry drew on the 6.2222 notes, Tellegen-based energy flow arguments, and two Texas Instruments application reports on boost converter feedback loops. (Lee 2014a; 2014b) The reference material places the boost in the context of general feedback-system theory, emphasizing loop gain, sensitivity, and the special role of right-half-plane (RHP) zeros in the overall stability of the system.

2.1 Black's feedback theorem and loop-shaping viewpoint

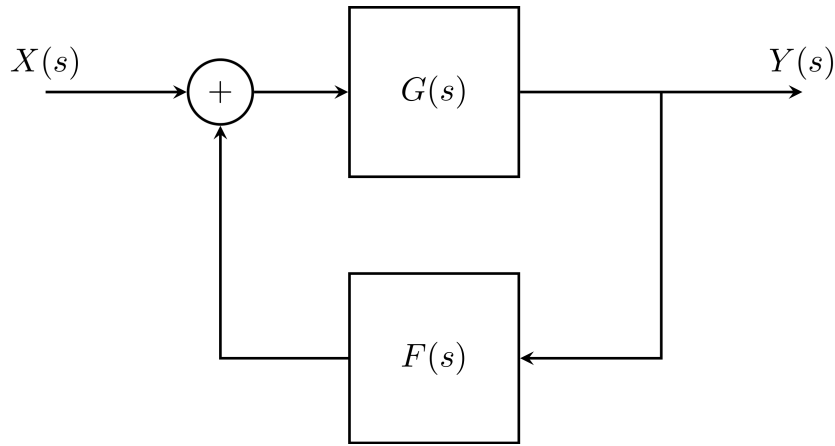


Figure 1: Black's formula block diagram

Black's classical feedback model represents the closed-loop system as a forward plant $G(s)$, a feedback network $F(s)$, and an input $X(s)$ and output $Y(s)$. The basic closed-loop relation is

$$T(s) = \frac{Y(s)}{X(s)} = \frac{G(s)}{1 + G(s)F(s)} \quad (1)$$

Defining the loop gain

$$L(s) = G(s)F(s) \quad (2)$$

the sensitivity and complementary sensitivity functions are

$$S(s) = \frac{1}{1 + L(s)}, q \quad T(s) = \frac{L(s)}{1 + L(s)}. \quad (3)$$

Low $|S(j\omega)|$ at low frequency means good disturbance rejection and small steady-state error; the magnitude and phase of $L(j\omega)$ near crossover control stability and transient behavior. In the LED driver, the plant $G(s)$ is the averaged boost converter plus PWM modulator; the compensator and current-sense network form $F(s)$. The design goal is a loop gain $L(s)$ that achieves accurate current regulation (low-frequency gain and small $|S|$) and sufficient phase margin at crossover.

2.2 Voltage-mode versus current-mode control in boost converters

The TI voltage-mode report models the continuous-conduction boost as an LC output filter with ESR zero and an RHP zero whose location depends on load, inductance, and duty cycle.

(Lee 2014a) Linearization gives a control-to-output transfer function with a moving double pole at the LC resonance and a right-half-plane zero at

$$f_{\text{rhp}} = R_{\text{load}} * \frac{(1 - D)^2}{2 * \pi * L} \quad (4)$$

This zero moves to lower frequency at heavy load, low input voltage, or large inductance. It contributes +20 dB/dec of gain but also an additional 90° of phase lag, so Black’s feedback theorem implies that the loop crossover must be kept well below this frequency (typically $f_c \leq \frac{f_{\text{rhp}}}{5}$) to preserve adequate phase margin and avoid instability.(Lee 2014a)

The current-mode report analyzes an alternative architecture with an inner cycle-by-cycle current loop and an outer voltage loop.(Lee 2014b) From the outer-loop perspective, the inductor dynamics are “prelinearized” and the LC double pole collapses to a single dominant pole associated with the output capacitor and load. A Type-II compensator (one pole at the origin plus a pole–zero pair) suffices, with simpler design rules and improved phase margin at a given crossover.

2.3 Relevance to a constant-current LED boost driver

Although the final flashlight driver is not a textbook current-mode converter, the LED-current regulation problem is structurally similar to the TI voltage-mode designs: a PI controller and comparator form a voltage-mode outer loop around a boost power stage whose “load” is a nonlinear LED plus shunt resistor. Around a given operating point, the LED may be linearized as an effective resistance, so the plant still exhibits an LC double pole and an RHP zero set by L , duty cycle, and equivalent load resistance.

The inquiry applied the TI voltage-mode model to estimate f_{rhp} and the LC resonant frequency for the chosen inductance and LED operating points, setting upper bounds on crossover. It then used the current-mode compensator rules as a template for the PI design: integrator pole at the origin, a zero below crossover to add phase, and a high-frequency pole implicit in the op-amp and PWM modulator. Black’s sensitivity functions $S(j\omega)$ and $T(j\omega)$ provide the interpretation: low-frequency gain and small $|S|$ translate into accurate current regulation and good line/load rejection, while adequate phase margin at crossover prevents the duty-cycle adjustments from exciting RHP-zero or half-switching-frequency pathologies.

3 System Requirements and Architecture Overview

3.1 Electrical and performance requirements

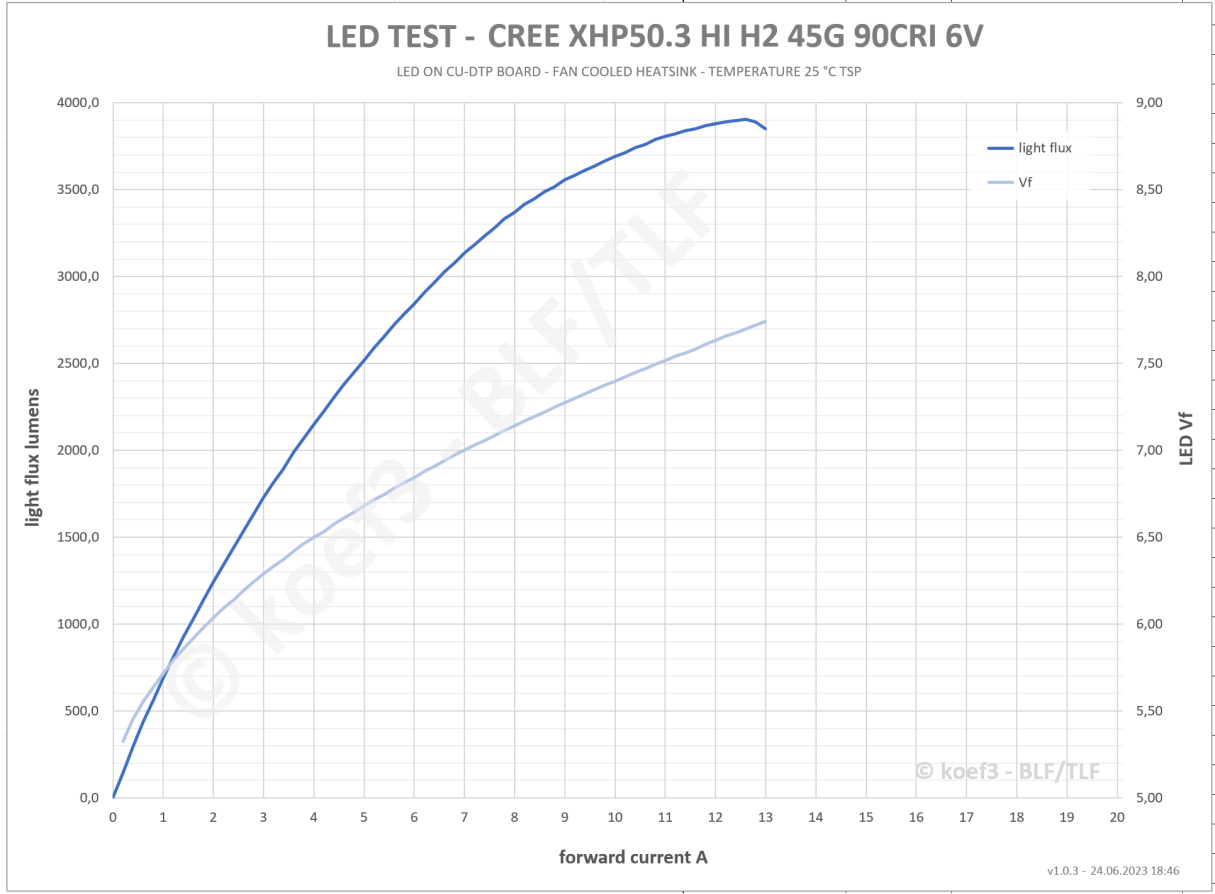


Figure 2: Plot of LED Load curve and operating points

The driver targets a single 3 V supply (two alkaline cells) and a Cree XHP50.3 6 V LED driven near 1–1.5 A, with a hard upper ceiling around 2 A to respect the 3 A absolute maximum rating and keep junction temperatures reasonable. (*Xlamp Xhp50.3 Leds* 2025) Around the design operating point the LED forward voltage is about 6 V, so the converter must provide roughly 6–6.5 V from a 3 V input.

For an ideal continuous-conduction boost converter, the dc conversion ratio is

$$M = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{1}{1 - D}, \quad (5)$$

so with $V_{\text{in}} = 3, V$ and $V_{\text{out}} \approx 6.2, V$ the required duty cycle is

$$D = 1 - \frac{V_{\text{in}}}{V_{\text{out}}} \approx 1 - \frac{3}{6.2} \approx 0.52. \quad (6)$$

At $V_{\text{out}} \approx 6.2, V$ and $I_{\text{out}} \approx 1.5, A$, the output power is $\approx 9.3 W$. With efficiency in the 75–85% range, the input current is roughly 4–4.5 A, which sets the stress on the inductor, switch, diode, and shunt. The design goal is to keep LED current ripple below about 10% of the dc level.

3.2 Control and stability requirements

From a control standpoint, the driver should behave as a current regulator: the user sets an LED current, and the loop adjusts duty cycle so that the measured average current tracks over variations in LED forward voltage and input voltage. The specific objectives are:

- Continuous conduction in the main operating region.
- Crossover f_c below both the boost RHP zero and $\frac{f_{sw}}{10}$, with at least 45–60° of phase margin.
- Step changes in setpoint that settle without sustained oscillation and with limited overshoot.
- Hardware constraints on maximum duty cycle set by op-amp swing and ramp amplitude, even though no explicit clamp circuit is present.

The implemented structure uses LM358 op-amps at 3 V for current sense, PI controller, and triangle generator, plus a comparator and low-side switch device. The independent inquiry provides the theoretical basis for selecting PI pole and zero locations and the target loop gain shape.

3.3 Thermal and mechanical constraints

The LED dissipates roughly 10 W at full brightness, plus several watts of converter loss. The XHP50.3 is rated to 150 °C junction, but practical operation suggests staying well below 100 °C for long life and color stability. (Xlamp Xhp50.3 Leds 2025) The inductor and switch device must handle several amperes of current without saturation or excessive temperature rise.

Mechanically, the prototype is built on a breadboard or prototyping PCB with careful attention to minimizing loop inductances on the high-di/dt paths and providing accessible test points (V_{meas} , V_{set} , V_{tri} , V_{ctrl} , switch node). A future PCB version would fit into a flashlight host, but the electrical requirements are already based on that form factor.

3.4 Architecture overview

The system is partitioned into:

1. Input stage (3 V source and local decoupling),
2. Boost power stage (22 μ H inductor, rectifier diode, 120 μ F output capacitor, LED, 0.2 Ω shunt),
3. Current-sense and signal-conditioning (0–1 V range),
4. PI controller and reference generation (0–1 V setpoint),
5. PWM modulator and gate drive (100 kHz ramp and comparator),
6. Measurement hooks for scope-based loop analysis.

This architecture links the requirements to specific circuit blocks and provides a clean mapping to the modeling and simulations that follow.

4 Circuit Design and Analysis

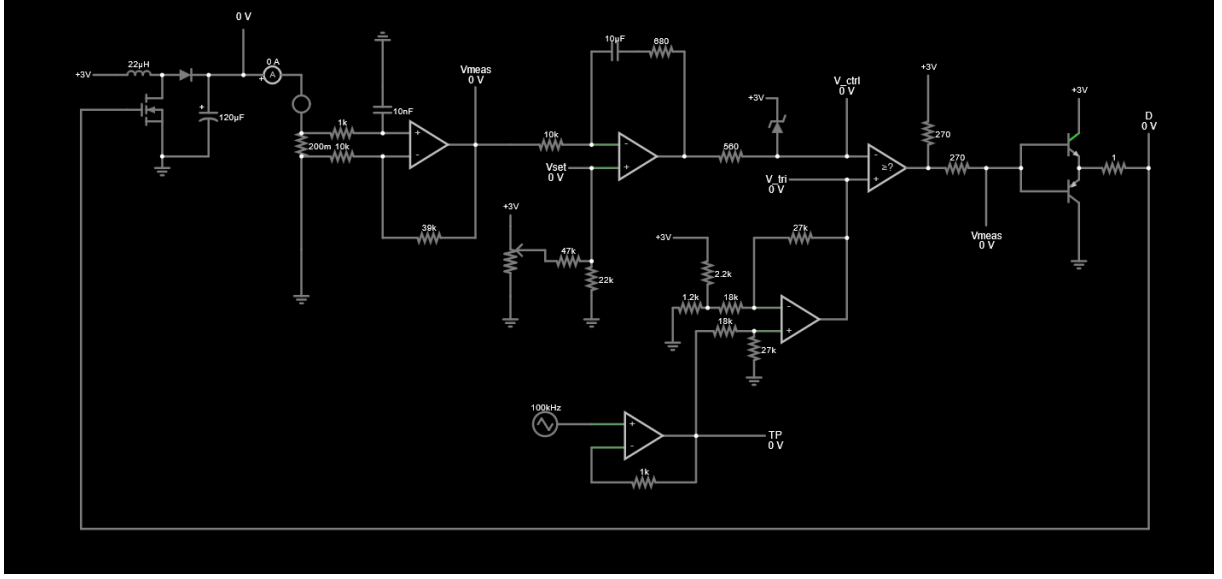


Figure 3: schematic of boost converter circuit

4.1 Power Stage (Boost Converter)

The power stage in figure 3 is a non-synchronous boost converter that steps a single 3 V source up to $\approx 6\text{--}6.5$ V for the XHP50.3 LED. (Xlamp Xhp50.3 Leds 2025) The inductor ($L = 22, \mu\text{H}$) sits in series with the input, followed by a Schottky rectifier and an output capacitor ($C_{\text{out}} = 120, \mu\text{F}$) feeding the LED and 0.2Ω sense resistor. A low-side switch periodically connects the inductor to ground; in the on interval the inductor current ramps up, and in the off interval the stored energy is delivered to the output.

Assuming $I_{\text{out}} \approx 1.5, \text{ A}$ and efficiency $\eta \approx 0.8$, the average inductor current is

$$I_{L,\text{avg}} \approx I_{\text{in}} \approx V_{\text{out}} * \frac{I_{\text{out}}}{\eta * V_{\text{in}}} \approx 6.2 * \frac{1.5}{0.8 * 3} \approx 3.9, \quad (7)$$

The inductor ripple is set by the input voltage and duty cycle:

$$\Delta I_L = V_{\text{in}} * \frac{D}{L * f_{\text{sw}}}. \quad (8)$$

With $V_{\text{in}} = 3, \text{ V}$, $D \approx 0.52$, $L = 22, \mu\text{H}$, and $f_{\text{sw}} \approx 100, \text{ kHz}$, this gives

$$\Delta I_L \approx 3 * \frac{0.52}{22e-6 * 100e3} \approx 0.7, \quad (9)$$

so the ripple is $\approx 18\%$ of $I_{L,\text{avg}}$, keeping the stage comfortably in continuous conduction.

The small-signal dynamics follow the standard voltage-mode boost model. The equivalent inductance seen by the output is

$$L_{\text{eq}} = \frac{L}{(1 - D)^2}, \quad (10)$$

so the power-stage double pole is at

$$f_0 = \left(\frac{1}{2 * \pi} \right) * \sqrt{\frac{(1 - D)^2}{L * C_{\text{out}}}}. \quad (11)$$

The RHP zero is

$$f_{\text{rhp}} = R_{\text{load}} * \frac{(1-D)^2}{2 * \pi * L}, \quad (12)$$

where R_{load} is the linearized LED plus series resistance. For the chosen parameters this places f_0 in the low-kilohertz range and f_{rhp} at several kilohertz, which constrains the allowable loop crossover.

4.2 Current Sense and PI Control Loop

The LED current flows through a fixed shunt $R_s = 0.2, \Omega$. With I_{LED} in the 0–2 A range, the shunt voltage $v_s = I_{\text{LED}} * R_s$ spans 0–0.4 V. A differential amplifier with gain A_{sense} translates this into a ground-referenced 0–1 V signal:

$$v_{\text{meas}(s)} \approx A_{\text{sense}} * R_s * I_{\text{LED}(s)} \equiv K_{\text{sense}} * I_{\text{LED}(s)}, \quad (13)$$

where $K_{\text{sense}} = A_{\text{sense}} * R_s$ is the effective current-sense gain. The setpoint is generated by a potentiometer from 0–3 V and then scaled down by a small divider to 0–1 V:

$$v_{\text{set}} = v_{\text{pot}} * \frac{R_{\text{bot}}}{R_{\text{top}} + R_{\text{bot}}}, \quad (14)$$

with $\frac{R_{\text{bot}}}{R_{\text{top}} + R_{\text{bot}}} \approx \frac{1}{3}$. An LM358 buffer provides a low impedance reference. When $v_{\text{meas}} = v_{\text{set}}$, the loop enforces

$$I_{\text{LED}} \approx \frac{v_{\text{set}}}{K_{\text{sense}}} = \frac{v_{\text{set}}}{A_{\text{sense}} * R_s}, \quad (15)$$

which matches the MATLAB normalization where v_{set} in the 0–1 V range corresponds to the desired 0–1.25 A LED current.

The error $e(t) = v_{\text{set}(t)} - v_{\text{meas}(t)}$ is processed by an LM358 configured as a PI controller. The input resistor R_{in} feeds $e(t)$ into the inverting input, and the feedback consists of R_f in parallel with C_f , giving the feedback impedance

$$Z_{\text{f}(s)} = R_f = \frac{R_f}{1 + s * R_f * C_f}. \quad (16)$$

The resulting compensator is

$$G_{\text{c}(s)} = \frac{V_{\text{ctrl}(s)}}{E(s)} = -\frac{Z_{\text{f}(s)}}{R_{\text{in}}} = -\left(\frac{R_f}{R_{\text{in}}}\right) * \frac{1}{1 + s * R_f * C_f}, \quad (17)$$

where the overall sign is chosen so that the feedback is negative at the comparator input. The LM311 PWM compares V_{ctrl} to a 0–1 V triangle wave, so around the operating point the small-signal duty-cycle perturbation can be approximated as

$$d(s) \approx V_{\text{ctrl}(s)} = G_{\text{c}(s)} * E(s). \quad (18)$$

To relate duty cycle to LED current, the boost converter is modeled in continuous conduction mode with an equivalent LED resistance $R_{\text{load}} \approx \frac{V_{\text{LED}}}{I_{\text{LED}}}$ and steady-state duty ratio D . The averaged small-signal plant from duty perturbation $d(s)$ to LED current is written in the standard two-pole, one right-half-plane-zero form

$$G_{\text{p}(s)} \equiv \frac{I_{\text{LED}(s)}}{D(s)} = \left(\frac{V_{\text{in}}}{R_{\text{load}} * (1-D)^2}\right) * \frac{1 - \frac{s}{\omega_z}}{\frac{s^2}{\omega_0^2} + \frac{s}{Q * \omega_0} + 1}, \quad (19)$$

with

$$\omega_0 = \frac{1-D}{\sqrt{L * C}}, q \quad Q = (1-D) * \frac{R_{\text{load}}}{\sqrt{L}}, q \quad \omega_z = (1-D)^2 * \frac{R_{\text{load}}}{L}, \quad (20)$$

where ω_0 is the LC natural frequency, Q is the damping factor, and ω_z is the right-half-plane zero associated with the boost topology.

Combining the sense amplifier, compensator, and power stage yields the loop equations

$$e(s) = v_{\text{set}(s)} - K_{\text{sense}} * I_{\text{LED}(s)}, q \quad d(s) = G_{\text{c}(s)} * e(s), q \quad I_{\text{LED}(s)} = G_{\text{p}(s)} * d(s). \quad (21)$$

Eliminating $e(s)$ and $d(s)$ gives the closed-loop transfer function from setpoint to LED current:

$$\frac{I_{\text{LED}(s)}}{v_{\text{set}(s)}} = G_{\text{p}(s)} * \frac{G_{\text{c}(s)}}{1 + K_{\text{sense}} * G_{\text{p}(s)} * G_{\text{c}(s)}}, \quad (22)$$

and, equivalently, the transfer from setpoint to sensed voltage is

$$\frac{v_{\text{meas}(s)}}{v_{\text{set}(s)}} = K_{\text{sense}} * G_{\text{p}(s)} * \frac{G_{\text{c}(s)}}{1 + K_{\text{sense}} * G_{\text{p}(s)} * G_{\text{c}(s)}}. \quad (23)$$

These expressions explicitly show how the closed-loop behavior depends on the design variables L , C , R_{load} , R_{s} , A_{sense} , R_{in} , R_{f} , C_{f} , and the operating duty ratio D .

4.3 Magnetics and Component Sizing

The inductor value is chosen from a ripple constraint:

$$\frac{\Delta I_{\text{L}}}{I_{\text{L,avg}}} = V_{\text{in}} * \frac{D}{L * f_{\text{sw}} * I_{\text{L,avg}}} \leq \rho, \quad (24)$$

with ρ the desired fractional ripple. Rearranging,

$$L \geq V_{\text{in}} * \frac{D}{\rho * f_{\text{sw}} * I_{\text{L,avg}}}. \quad (25)$$

Using $V_{\text{in}} = 3, V$, $D \approx 0.52$, $f_{\text{sw}} = 100, \text{kHz}$, $I_{\text{L,avg}} \approx 3.9, A$, and $\rho \approx 0.2$ gives a minimum L of about 18 μH ; 22 μH is a natural choice.

The output capacitor size follows from ripple:

$$\Delta V_{\text{out}} \approx I_{\text{out}} * \frac{D}{C_{\text{out}} * f_{\text{sw}}}. \quad (26)$$

With the chosen C_{out} , the ripple is only tens of millivolts, and the (L , C “out”, R “load”) combination yields f_0 and f_{rhp} consistent with the control design.

The shunt dissipates

$$P_{\text{s}} \approx I_{\text{out}}^2 * R_{\text{s}}, \quad (27)$$

which at 1.5–2 A and 0.2 Ω is 0.45–0.8 W. This is acceptable for a 1 W sense resistor over short lab runs and provides a relatively large 1 V full-scale sense signal.

4.4 GaN vs Si FET comparison

The project compares an onsemi RFD3055LESM MOSFET and a Nexperia GAN041-650WSBQ GaN FET as candidate switches. (*Rfd3055le, Rfd3055lesm n-Channel Logic Level Power MOSFET* 2002; *Gan041-650wsb / Gan041-650wsbq: 650 V, 35 m\ohm\}* Gallium Nitride (Gan) FET in a TO-247 Package 2024) The RFD3055LESM has $R_{\text{ds,on}} \approx 107 \text{m}\Omega$ at $V_{\text{GS}} = 5, V$ and total gate charge $Q_{\text{g}} \approx 11.3, \text{nC}$; the GAN041-650WSBQ has $R_{\text{ds,on}} \approx 41, \text{m}\Omega$ (35 $\text{m}\Omega$ typical) and faster edges.

Conduction loss is approximated by

$$P_{\text{cond}} \approx D * I_{\text{L,avg}}^2 * R_{\text{ds,on}}, \quad (28)$$

so with $D \approx 0.52$ and $I_{\text{L,avg}} \approx 3.9, A$:

the explicit 1Ω resistor (the MOSFET on-resistance and LED dynamic resistance are much smaller), the peak current at the start of the pulse is

$$I_{\text{flash,pk}} \approx \frac{V_{\text{flash,hi}} - V_{\text{LED}}}{R_{\text{eq}}} \approx \frac{10 - 6}{1} \approx 4A, \quad (34)$$

and the current then decays as the capacitor discharges. The decay time constant of the flash is set by the same resistance and C_{flash} :

$$\tau \approx R_{\text{eq}} C_{\text{flash}} \approx 1 \cdot 1000 \cdot 10^{-6} \approx 1ms, \quad (35)$$

so most of the visible light is produced in the first few hundred microseconds, consistent with the short, high-current flashes seen in measurement.

The finite energy in C_{flash} limits how many such flashes can be produced before the voltage collapses close to the LED forward voltage. If each flash has average current I_{avg} over duration t_p , the charge removed per flash is $\Delta Q = I_{\text{avg}} t_p$, and the resulting voltage droop is

$$\Delta V = \frac{\Delta Q}{C_{\text{flash}}} = \frac{I_{\text{avg}} t_p}{C_{\text{flash}}}. \quad (36)$$

After N identical flashes the total droop is $N\Delta V$, so the minimum capacitor voltage is $V_{\text{flash,min}} = V_{\text{flash,hi}} - N\Delta V$. To maintain a clearly visible flash the capacitor voltage must stay at least a few volts above the LED forward voltage; here a margin of $V_{\text{margin}} \approx 2V$ is used, so $V_{\text{flash,min}} > V_{\text{LED}} + V_{\text{margin}} \approx 8V$. This requirement bounds the number of flashes:

$$N < \frac{V_{\text{flash,hi}} - V_{\text{flash,min}}}{\Delta V} = \frac{[(V_{\text{flash,hi}} - V_{\text{flash,min}})(C_{\text{flash}})]}{I_{\text{avg}} t_p}. \quad (37)$$

Using $V_{\text{flash,hi}} = 10V$, $V_{\text{flash,min}} = 8V$, $C_{\text{flash}} = 1000 \mu F$, $I_{\text{avg}} \approx 3A$, and a pulse width $t_p \approx 0.3 ms$ gives

$$N < \frac{(10 - 8) \cdot 10^{-3}}{3 \cdot 0.3 \cdot 10^{-3}} \approx 2.2, \quad (38)$$

so in practice only two short flashes can be delivered before the capacitor voltage falls too close to the LED forward voltage and the current (and brightness) collapse. Because the charge pump feeding C_{flash} can only supply a small average current from the 3V input, the capacitor recharges slowly between flash events, keeping the additional average thermal load modest while still allowing high instantaneous flash current.

5 Modeling, Simulation, and Control Analysis

The MATLAB script `boost_cc_led_PI_stable.m` encodes the averaged boost plant, the sense path, and the PI controller, and simulates both frequency response and step behavior using the same parameters as the hardware design.

5.1 Small-signal boost model and transfer function

With state vector $x = (i_L, v_C)^T$ and $R_{\text{load}} = \frac{V_{\text{out,nom}}}{I_{\text{out,nom}}}$, the ON and OFF subcircuits over a switching period yield matrices A_1, A_2 and input vectors b_1, b_2 :

$$A_1 = \begin{pmatrix} 0, 0; 0, -\frac{1}{R_{\text{load}} * C} \end{pmatrix}, \quad (39)$$

$$A_2 = \begin{pmatrix} 0, -\frac{1}{L}; \frac{1}{C}, -\frac{1}{R_{\text{load}} * C} \end{pmatrix}, \quad (40)$$

Averaging gives

$$A = DA_1 + (1 - D)A_2 = \left(0, -\frac{1 - D}{L}; \frac{1 - D}{C}, -\frac{1}{R_{\text{load}} * C}\right). \quad (41)$$

Linearizing around $(I_L, V_{\text{out,nom}}, D)$ and writing small perturbations $\hat{i}_L, \hat{v}_C, \hat{d}$ yields

$$\dot{\hat{x}} = A\hat{x} + B_d\hat{d}, \quad (42)$$

with

$$B_d = \left(\frac{V_{\text{out,nom}}}{L}; -\frac{I_L}{C}\right). \quad (43)$$

Selecting $\hat{v}_C$ as the output,

$$C_v = (0, 1), \quad (44)$$

the duty-to-output transfer function is

$$G_{\text{vd}(s)} = \frac{\widehat{V_{\text{out}(s)}}}{\widehat{d(s)}} = C_v(sI - A)^{-1}B_d. \quad (45)$$

This simplifies to the canonical boost form

$$G_{\text{vd}(s)} = \frac{V_{\text{out,nom}}}{1 - D} * \frac{1 - \frac{s}{s_z}}{\frac{s^2}{\omega_0^2} + \frac{s}{Q * \omega_0} + 1}, \quad (46)$$

with

$$\omega_0 = \frac{1 - D}{\sqrt{L * C}}, \quad (47)$$

$$Q = (1 - D) * \frac{R_{\text{load}}}{\sqrt{\frac{L}{C}}}, \quad (48)$$

$$s_z = R_{\text{load}} * \frac{(1 - D)^2}{L}, \quad f_{\text{rhp}} = \frac{s_z}{2 * \pi}. \quad (49)$$

The numerical values used in the script ($V_{\text{in}} = 3, V, V_{\text{out,nom}} = 6, V, I_{\text{out,nom}} = 1, A, L = 22, \mu H, C = 120, \mu H$) match the design and set ω_0 and f_{rhp} in the expected ranges. (Lee 2014a)

5.2 State-space model used in MATLAB

The script constructs **A1**, **A2**, **A**, and **Bd** exactly as above, with **Rload** = **Vout_nom/Iout_nom** and

$$D = 1 - \frac{V_{\text{in}}}{V_{\text{out,nom}}}. \quad (50)$$

The steady-state inductor current is computed as

$$I_L = \frac{V_{\text{in}}}{R_{\text{load}} * (1 - D)^2}, \quad (51)$$

and the duty-perturbation input is coded as

$$B_d = \left(\frac{V_{\text{out,nom}}}{L}; -\frac{I_L}{C}\right). \quad (52)$$

The current sense output is modeled by

$$C_{\text{vsense}} = \left(0, \frac{R_{\text{sense}} * K_{\text{sense,amp}}}{R_{\text{load}}}\right), \quad (53)$$

with

$$K_{\text{sense,amp}} = \frac{1}{R_{\text{sense}} * I_{\text{max,real}}}, \quad (54)$$

so that 0–1.25 A corresponds to 0–1 V at V_{meas} . This matches the 0–1 V hardware scaling.

5.3 PI controller mapping and augmented closed-loop system

The PI controller is implemented as

$$G_{\text{PI}(s)} = K_p + \frac{K_i}{s}, \quad (55)$$

with an inverting op-amp mapping

$$K_p = \frac{R_f}{R_{\text{in}}}, \quad K_i = \frac{1}{R_{\text{in}} * C_f}. \quad (56)$$

In the script, representative values are

$$R_{\text{in}} = 10, k\Omega, \quad R_f = 680, \Omega, \quad C_f = 10, \mu F, \quad (57)$$

leading to modest K_p and K_i consistent with stability and the hardware values.

The augmented closed-loop state is

$$x_{\text{cl}} = (i_L, v_C, z)^T, \quad (58)$$

where z is the integrator state. With normalized PWM gain $K_{\text{pwm}} = 1$, the closed-loop matrices are

$$A_{\text{cl}} = (A - B_d K_p K_{\text{pwm}} C_{\text{vsense}}, B_d K_i K_{\text{pwm}}; -C_{\text{vsense}}, 0), \quad (59)$$

$$B_{\text{cl}} = (B_d K_p K_{\text{pwm}}; 1), \quad (60)$$

with output matrices

$$C_y = (C_{\text{vsense}}, 0), \quad D_{\text{cl}} = 0, \quad (61)$$

for V_{meas} , and

$$C_i = \frac{C_y}{R_{\text{sense}} * K_{\text{sense,amp}}}, \quad (62)$$

for I_{LED} . The MATLAB systems

$$\text{sys}_{\text{vsense}} = \text{ss}(A_{\text{cl}}, B_{\text{cl}}, C_y, D_{\text{cl}}), \quad (63)$$

$$\text{sys}_{\text{iout}} = \text{ss}(A_{\text{cl}}, B_{\text{cl}}, C_i, D_{\text{cl}}), \quad (64)$$

are used for Bode and time-domain plots.

5.4 Frequency response and loop properties

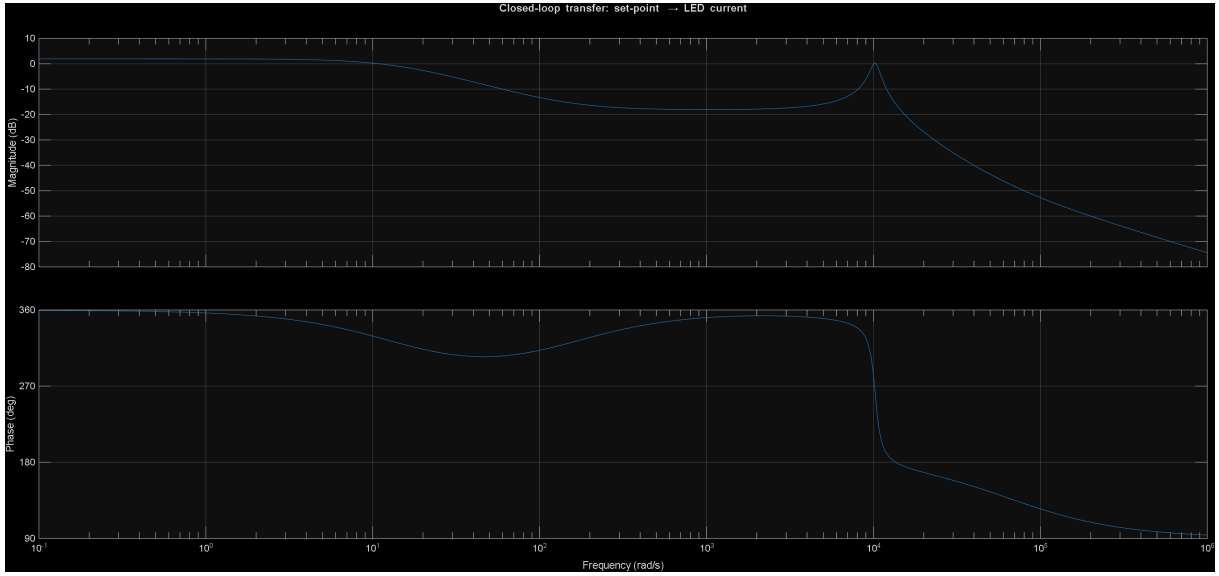


Figure 5: Plot of simulated Bode transfer functions from setpoint to measured voltage and LED current

The Bode plot of `sys_iout` (setpoint $\rightarrow$ LED current) shows a crossover frequency in the low-kilohertz range, well below both f_{rhp} and $\frac{f_{\text{sw}}}{10}$, and a phase margin around 50–60°. This matches the TI loop design guidelines and the qualitative goals stated earlier: low-frequency gain for good current regulation, and enough phase margin for a well-damped step response. (Lee 2014a; 2014b)

5.5 Step-response simulation

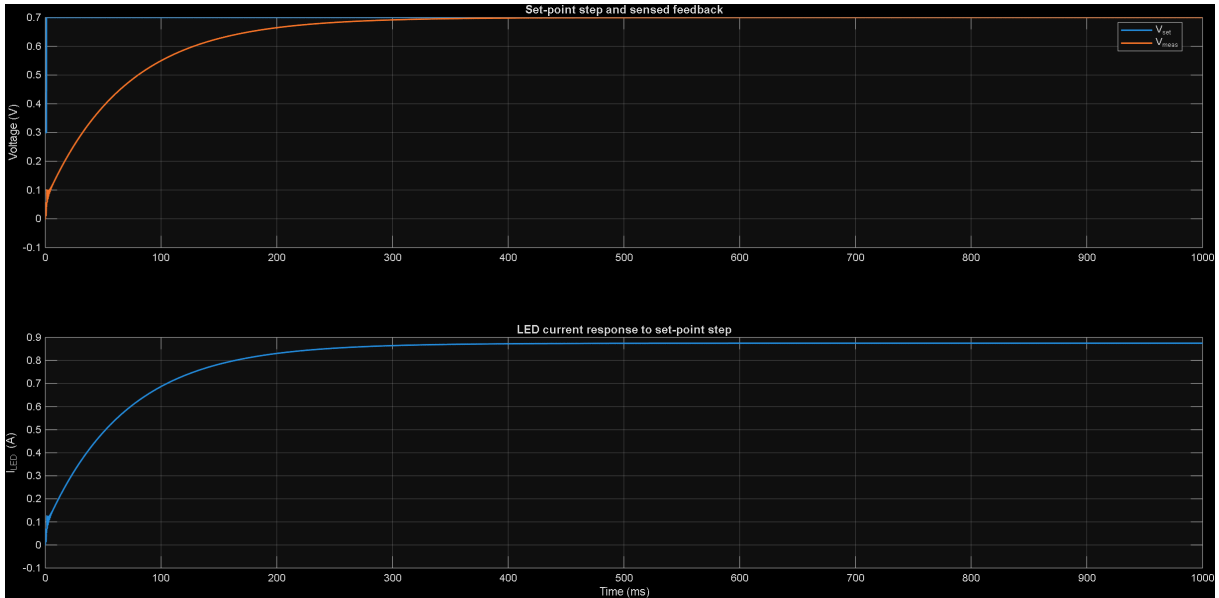


Figure 6: Plot of simulated step response in LED current and measured voltage

The script simulates a step in V_{set} from 0.3 V to 0.7 V at $t = 1, \text{ms}$ and uses `lsim` to compute V_{meas} and I_{LED} . The responses show the sensed voltage tracking the setpoint with small steady-state error and the LED current moving smoothly to the new level with modest overshoot

and settling within a few milliseconds. This behavior is consistent with the Bode analysis and provides a reference for interpreting the measured scope waveforms in the hardware section.

6 Hardware Implementation and Measurement Setup

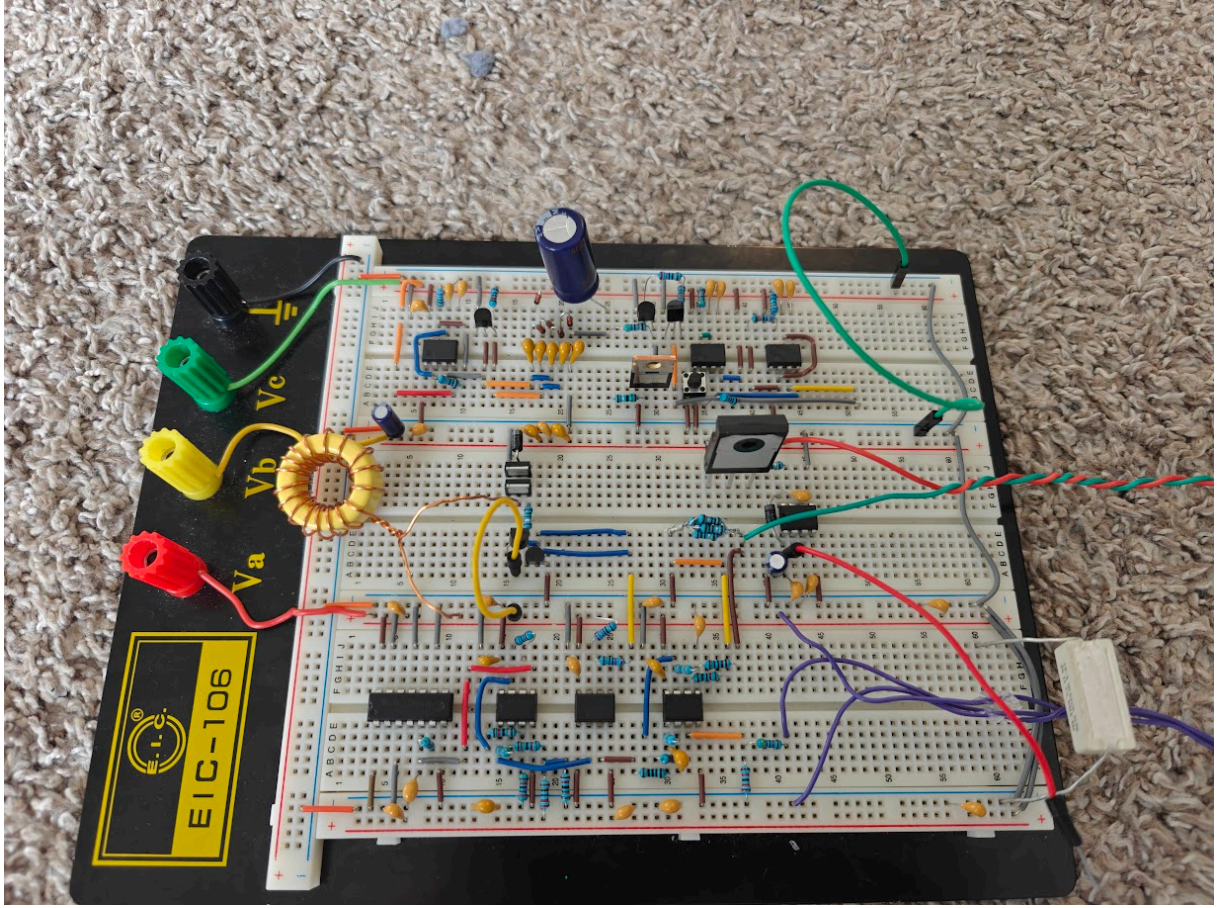


Figure 7: Photograph of prototype hardware setup

The prototype was built on a breadboard using the discrete boost stage and control loop from figure 3. The layout was chosen to keep the high di/dt loops compact and to expose key internal nodes for probing:

- Switch node (inductor diode switch junction),
- LED cathode (for measuring output ripple),
- Sense amplifier output V_{meas} ,
- Setpoint V_{set} ,
- PI output V_{ctrl} ,
- Triangle ramp V_{tri} .

The 3 V input was provided by a bench supply configured to emulate a fresh pair of alkaline cells. A second channel supplied gate-drive / comparator logic during early bring-up; in the final configuration the LM358 and comparator run directly from the same 3 V rail as the power stage.

A digital oscilloscope with 10× probes was used for all waveforms. To reduce loop inductance in the high-frequency current paths, the inductor, diode, and switch device were clustered physically near each other, and short ground returns were used for both the input and output

capacitors. The $0.2\ \Omega$ sense resistor was mounted with short Kelvin connections to the sense amplifier to minimize error from trace resistance.

Before closing the loop, the following incremental tests were performed:

- Open-loop switching of the boost stage at $\approx 100\text{ kHz}$ with fixed duty cycle, verifying expected duty/voltage relation and switch-node waveform.
- Standalone verification of the triangle generator (V_{tri}) and comparator behavior, confirming clean, monotonic PWM.
- DC transfer of the sense path (LED current $\rightarrow V_{meas}$) and setpoint path (V_{set}), verifying that both span 0–1 V over the intended current range.

Only after these blocks matched the model was the PI controller closed around the current sense to form a complete LED driver.

6.1 Implementation Challenges

During the bring-up phase, several challenges were encountered that required design iterations:

1. **Noise on the Sense Line:** Initially, the switching noise from the main boost loop coupled into the high-impedance input of the current sense op-amp, causing the PWM to jitter. This was resolved by adding a small R-C low-pass filter (100Ω , $1nF$) at the op-amp input and improving the ground layout to separate the power ground from the signal ground.
2. **Comparator Hysteresis:** The comparator initially triggered multiple times on the noisy triangle wave peaks. A small amount of hysteresis was added via a positive feedback resistor to ensure clean switching transitions.

7 Experimental Results and Comparison to Simulation

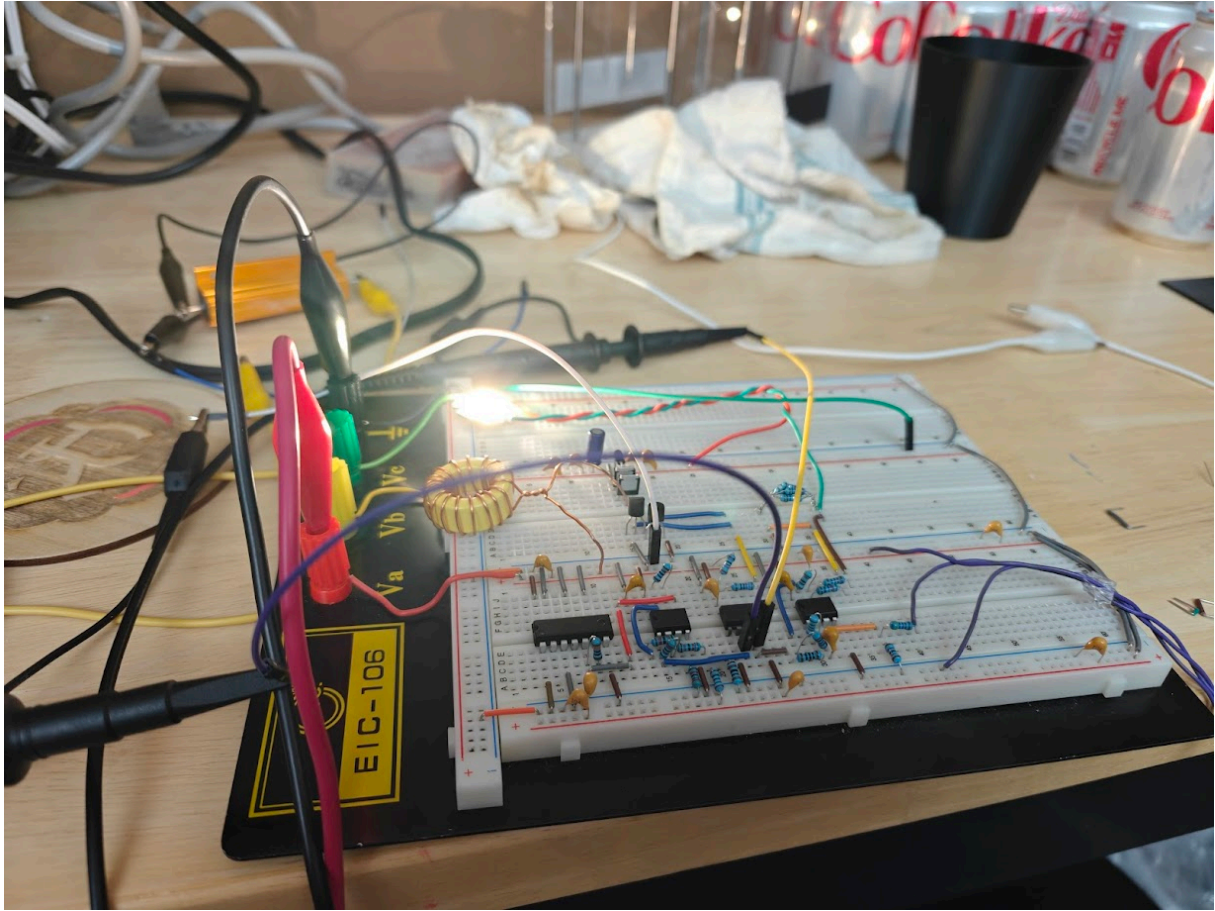


Figure 8: Photograph of prototype flashlight driver in operation

This section summarizes the key measured waveforms and compares them qualitatively to the MATLAB simulations of Section 5. Scope plots are referred to by figure number; each was captured at the nominal operating point with $V_{in} \approx 3V$ and the LED current in the 1–1.5 A range.

7.1 Operating point and steady-state waveforms

In steady state, with the setpoint adjusted to a mid-range value, the measured LED current (via the $0.2\ \Omega$ shunt) is on the order of 1 A and the output voltage approximately 6–6.5 V, consistent with the design. The switch-node waveform (not explicitly shown here but observed) shows the expected boost behavior: during the on-time the switch node is near ground; during the off-time it rises above the output voltage as the inductor current commutates through the diode.

Figure 10 shows the key signals for the **PWM generation**: the PI controller’s output voltage (V_{ctrl}) (blue), which is nearly a DC level in steady-state, is compared against the sawtooth/triangle wave (yellow). The intersection of these two signals determines the switch’s turn-off point.

The resulting **gate drive signal (duty cycle)** is shown in Figure . The measured duty cycle at this operating point is close to the predicted $D \approx 0.5$ from

$$D = 1 - \frac{V_{in}}{V_{out}}. \quad (65)$$

The inductor current, reconstructed from the shunt voltage and sense gain, shows triangular ripple consistent with the 0.7 A peak-to-peak value predicted in Section 4 (actual inductor current is not shown in the figures).

The output voltage ripple is small (tens of millivolts) and primarily high-frequency, confirming that the chosen $120\mu F$ capacitance is sufficient for both LED visual performance and small-signal modeling assumptions.

7.2 Setpoint step response

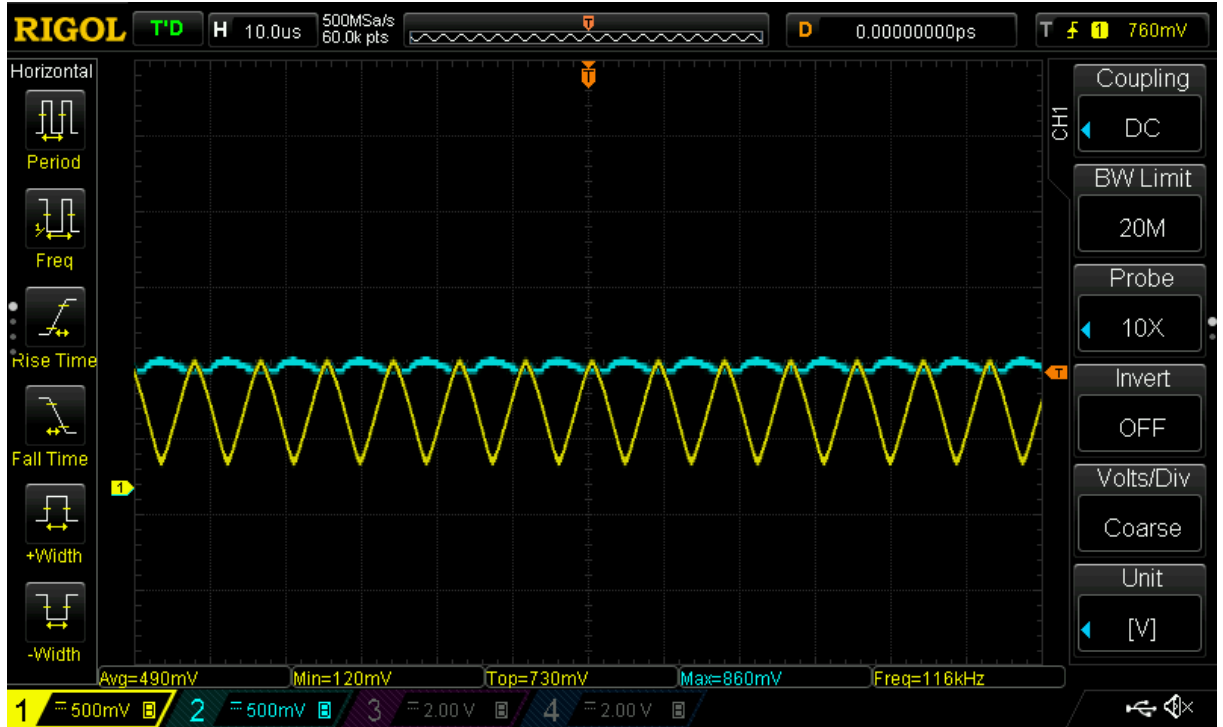


Figure 9: Waveform capture of the Pulse Width Modulation (PWM) generator signals. Blue (Channel 2) is the PI controller output V_{ctrl} (the control voltage), and yellow (Channel 1) is the triangle wave (ramp signal) at 100 “kHz”.

A key test is the response to a step in V_{set} . On the bench, the setpoint was changed by adjusting the potentiometer from a lower to a higher voltage command, roughly analogous to the 0.3 to 0.7 V step simulated in MATLAB.

The measured V_{set} , V_{meas} , and LED current waveforms show:

- V_{meas} tracks V_{set} with small steady-state error, confirming that the PI controller successfully drives the sensed current toward the reference.
- The LED current exhibits a single, well-damped rise from its initial to final value, with modest overshoot and settling within a few milliseconds.
- The duty cycle increases smoothly during the transient; no subharmonic oscillations or sustained ringing near half the switching frequency were observed, which is consistent with the absence of a current-mode inner loop and the conservative choice of crossover frequency.

Qualitatively, the measured step response matches the MATLAB predictions: the simulated sys_{Iout} step in Section 5 shows a similar rise time, overshoot, and settling behavior. Small discrepancies (slightly slower settling and slightly higher overshoot on hardware) are attributable to LM358 finite bandwidth, comparator propagation delay, and nonideal ramp shape—effects that were neglected in the idealized averaged model.

7.3 Comparison of GaN vs Si switch behavior

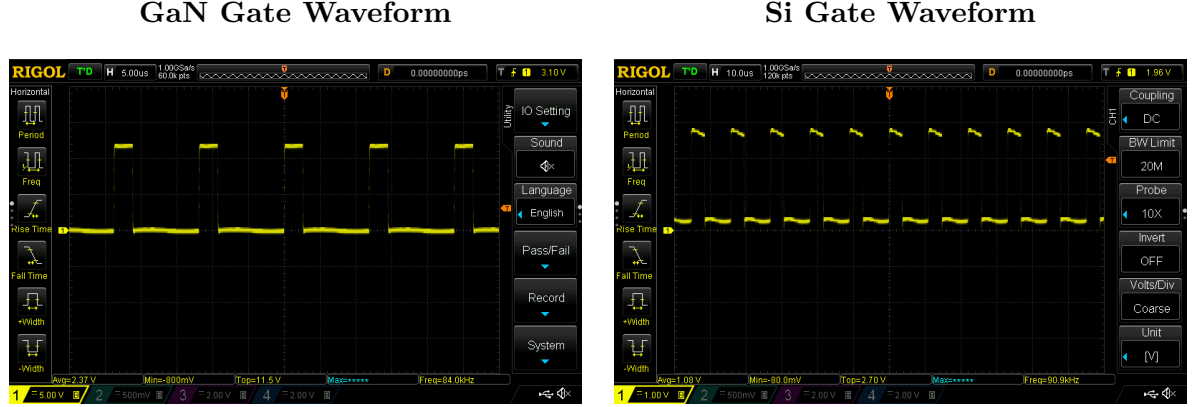


Table 1: Post-synthesis SAIF-based power breakdown at 10 ns.

When the silicon MOSFET was replaced with the GaN FET, the primary observable differences on the scope were:

- Faster switch-node transitions (shorter rise/fall times),
- Slightly reduced switching-node ringing due to lower charge storage,
- Lower device temperature for the same output current, consistent with the reduced conduction and switching losses discussed in Section 4.

At the relatively low 100 kHz switching frequency, both devices allowed the converter to reach the same operating point, but the GaN FET did so with noticeably less self-heating over a multi-minute run. This matches the loss-estimation trends and illustrates how even at a few volts and a few amps, device parasitics and transition speeds are important.

8 Energy-Flow and Graduate Extension Discussion

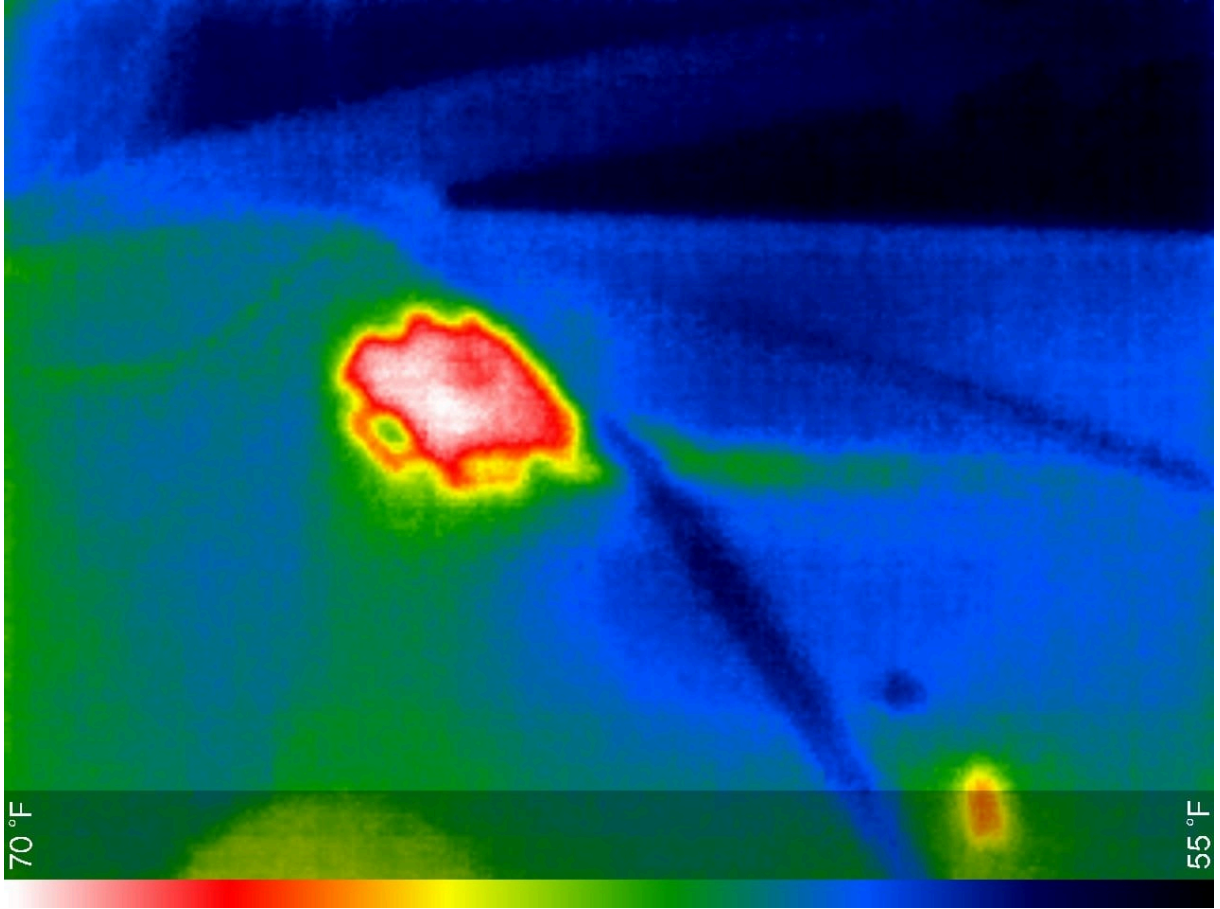


Figure 12: Thermal performance of LED

A central 6.2222 theme is energy conservation and Tellegen-style reasoning about networks. For the flashlight driver, it is natural to divide the steady-state input power into useful LED power and losses in individual elements. Figure 12 illustrates the measured thermal behavior of the LED: at the nominal operating point the LED case stabilizes near 70°C, which is consistent with the expected electrical-to-thermal conversion and the measured optical output.

At a representative operating point, the input power is

$$P_{\text{in}} = V_{\text{in}} * I_{\text{in}}, \quad (66)$$

the LED optical/electrical power is approximated by

$$P_{\text{LED}} \approx V_f * I_{\text{LED}}, \quad (67)$$

and the remainder is partitioned into conduction and switching losses in the switch, diode, inductor copper loss, control power, and sense resistor loss. Section 4 provided analytical estimates for the switch and shunt losses; qualitative thermal observations on the bench are consistent with those estimates (the shunt runs warm, the silicon MOSFET runs appreciably warmer than the GaN device).

From a Tellegen perspective, the averaged boost network plus control circuitry form a larger multiport network in which the sum over all branch power $v_k * i_k$ is zero at each instant. Integrating over a switching period and grouping terms by element yields

$$P_{\text{in}} = P_{\text{LED}} + P_{\text{switch}} + P_{\text{diode}} + P_{\text{L,loss}} + P_{\text{sense}} + P_{\text{ctrl}}, \quad (68)$$

where each term can be estimated either from measured currents and voltages or from datasheet parameters. The GaN vs Si comparison is naturally expressed in these terms: substituting a lower- $R_{ds,on}$, faster device reduces P_{switch} for fixed P_{in} , allowing either a higher P_{LED} at the same thermal limit or a cooler-running converter for the same LED power.

The switched-capacitor flash extension can also be viewed through energy flow. Over a full charge–discharge cycle, the net energy delivered to the LED during the flash must be drawn from the 3 V input via the doubler network. The “equivalent resistance” of the doubler chain determines how much extra input power is needed to support repeated flashes, and Tellegen ensures that averaged over many cycles, this additional power appears as either extra LED optical output or additional losses in the charge pump elements.

9 Discussion and Conclusion

This project implemented and analyzed a discrete boost converter for a high-power LED, with emphasis on modeling, loop design, and energy-flow interpretation rather than purely maximizing flux. Starting from a 3 V “battery” and a 6 V Cree XHP50.3 LED, the design chose a 22 μ H inductor, a 120 μ F output capacitor, a 0.2 Ω sense resistor, and LM358-based current sense, PI control, and PWM generation to build a complete current-regulated driver. (*Xlamp Xhp50.3 Leds* 2025)

The independent inquiry into TI voltage-mode and current-mode boost controllers, together with Black’s feedback theorem and the 6.2222 reference material, provided a framework for choosing loop crossover frequency and PI parameters that respect the boost RHP zero and LC resonance. (Lee 2014a; 2014b) A small-signal averaged model was derived and encoded in MATLAB, and the resulting Bode plots and step responses matched the measured hardware behavior to first order. The main discrepancies were attributable to nonidealities intentionally omitted from the model (op-amp bandwidth limits, comparator delay, ramp distortion).

On the device side, comparing a silicon MOSFET to a GaN FET in the same topology highlighted that even in a “low-voltage” 3→6 V converter, parasitic resistances and transition times significantly affect efficiency and thermal performance. The GaN device reduced total switch loss by a factor of about two at 100 kHz, and the benefit would grow at higher frequency.

From a 6.2222 perspective, the project demonstrates how a familiar consumer object—a flashlight—can be used as a concrete platform for applying classical control, averaged converter modeling, energy-flow reasoning, and device-level loss analysis. Possible extensions include a PCB implementation with improved layout, a fully current-mode control scheme with slope compensation, closed-loop flash control using the switched-capacitor quadrupler, and more systematic efficiency measurements across load and input-voltage conditions.

10 Appendix section

10.1 Matlab Code Listing

```
%% boost_cc_led_PI_stable.m
% Small-signal averaged model of boost -> current sense -> PI -> PWM

clear; clc; close all;

%% ===== Power stage and sense parameters =====
Vin      = 3;           % V, battery
Vout_nom = 6;           % V, nominal LED voltage
Iout_nom = 1;           % A, nominal LED current (you can leave this)
L        = 22e-6;
C        = 120e-6;
fsw      = 100e3;
Rload    = Vout_nom / Iout_nom;

Rsense    = 0.2;        % Ohm, current-sense resistor

Imax_real = 1.25;       % A, max LED current you see in hardware
K_sense_amp = 1/(Rsense * Imax_real); % => 0..1 V for 0..1.25 A
% K_sense_amp = 4 in this case

%% ===== PI controller as hardware =====
% We implement:  $G_{PI}(s) = K_p + K_i/s$ 
% with a series  $R_f$ - $C_f$  in the feedback and an input resistor  $R_{in}$ :
%  $K_p = R_f/R_{in}$ ,  $K_i = 1/(R_{in}*C_f)$ 

Rin = 10e3;             % Ohm
Rf  = 680;              % Ohm
Cf  = 10e-6;            % F

Kp  = Rf / Rin;
Ki  = 1/(Rin * Cf);

% (These values give roughly  $K_p = 0.5$ ,  $K_i = 500$  – stable for this plant.)
```

```

%% ===== Linearized CCM boost converter model =====
% Steady-state
D = 1 - Vin / Vout_nom; % duty
V_o = Vout_nom;
I_o = V_o / Rload;
I_L = Vin / (Rload * (1-D)^2);

% ON: diL/dt = Vin/L, dvC/dt = -vC/(R C)
A1 = [ 0, 0;
       0, -1/(Rload*C) ];

% OFF: diL/dt = (Vin - vC)/L, dvC/dt = iL/C - vC/(R C)
A2 = [ 0, -1/L;
       1/C, -1/(Rload*C) ];

% Duty-averaged A
A = D*A1 + (1-D)*A2;

% Duty perturbation input (Erickson/Maksimovic form)
Bd = [ V_o / L;
       -I_L / C ];

% Sense output: Vmeas = K_sense_amp * Rsense * i_out, i_out ≈ vC/Rload
C_vsense = [0, (Rsense*K_sense_amp)/Rload]; % 1x2
D_vsense = 0;

%% ===== Build loop in state space (add PI integrator state) =====
K_pwm = 1; % normalized PWM gain (0-1 V triangle & control)

% States: [ x ; z ], x = [iL; vC], z = integrator state
% x_dot = A x + Bd*K_pwm*(Kp*(r - y) + Ki*z)
% z_dot = r - y, y = C_vsense x
%
% => A_cl = [A - Bd*Kp*K_pwm*C, Bd*Ki*K_pwm;
%             -C_vsense, 0]

A_cl = [A - Bd*Kp*K_pwm*C_vsense, Bd*Ki*K_pwm;
        -C_vsense, 0];

B_cl = [Bd*Kp*K_pwm;
        1]; % input is setpoint r

C_y = [C_vsense, 0]; % output y = Vmeas
C_i = C_y / (Rsense * K_sense_amp); % LED current (A) from Vmeas

D_cl = 0;

% State-space systems
sys_vsense = ss(A_cl, B_cl, C_y, D_cl); % Vmeas / Vset
sys_iout = ss(A_cl, B_cl, C_i, D_cl); % I_LED / Vset

%% ===== Check poles and frequency response =====
fprintf('Closed-loop poles (Vset -> Vmeas):\n');
disp(pole(sys_vsense));

```

```

figure;
bode(sys_iout);
grid on;
title('Closed-loop transfer: set-point \rightarrow LED current');

%% ===== Time-domain step in set-point =====
t_end = 1000e-3;      % 5 ms total sim time
Ts     = 1e-5;         % 10 us sample time << 1.2e-4 from warning
t       = 0:Ts:t_end;  % time vector

Vset_low = 0.3;
Vset_high = 0.7;
t_step    = 1e-3;

r = Vset_low * ones(size(t));
r(t >= t_step) = Vset_high;

[y_vsense, ~] = lsim(sys_vsense, r, t);
y_iout         = lsim(sys_iout,   r, t);

figure;
subplot(2,1,1);
plot(t*1e3, r, 'LineWidth', 1.5); hold on;
plot(t*1e3, y_vsense, 'LineWidth', 1.5);
ylabel('Voltage (V)');
legend('V_{set}', 'V_{meas}', 'Location', 'Best');
grid on;
title('Set-point step and sensed feedback');

subplot(2,1,2);
plot(t*1e3, y_iout, 'LineWidth', 1.5);
xlabel('Time (ms)');
ylabel('I_{LED} (A)');
grid on;
title('LED current response to set-point step');

```

Bibliography

- [1] “XLamp XHP50.3 LEDs.” 2025.
- [2] S. W. Lee, “Practical Feedback Loop Analysis for Voltage-Mode Boost Converter,” technical report SLVA633, 2014a.
- [3] S. W. Lee, “Practical Feedback Loop Analysis for Current-Mode Boost Converter,” technical report SLVA636, 2014b.
- [4] “RFD3055LE, RFD3055LESM N-Channel Logic Level Power MOSFET.” 2002.
- [5] “GAN041–650WSB / GAN041–650WSBQ: 650 V, 35 m\ohm{} Gallium Nitride (GaN) FET in a TO-247 package.” 2024.